

Health Impact Report

Including health impacts in transport modelling and economic appraisal



A partnership project between
RMIT University and the
Victorian Department of Transport

Acknowledgements

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The *Transport Health Assessment Tool for Melbourne (THAT-Melbourne)* is available online at the [Australian Urban Observatory](#).

Enquiries regarding this report may be directed to:

Building 8, Level 11 RMIT University City campus
124 La Trobe Street Melbourne VIC 3000
E hlc@rmit.edu.au P 03 9925 4577

Our funding and research partners



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Abbreviations

ABS	Australian Bureau of Statistics	PIA	Planning Institute of Australia
ATAP	Australian Transport Assessment and Planning	NHMRC	National Health and Medical Research Council
AUO	Australian Urban Observatory	TAPPC	The Australian Prevention Partnership Centre
DOT	Victorian Department of Transport	THAT-Melbourne	Transport Health Assessment Tool Melbourne
EIT	European Institute of Innovation and Technology	RMIT	RMIT University
HEAT	Health Economic Assessment Tool	UKRI	United Kingdom Research Institute
HIA	Health Impact Assessment	VISTA	Victorian Integrated Survey of Travel and Activity
ITHIM	Integrated Transport and Health Impact Modelling Tool		



Executive Summary

This document is the final report for a two-year project titled *Including health impacts in transport modelling and economic appraisal*. The research was funded by RMIT University's Urban Futures Enabling Capability Platform Opportunity Fund for Research and Translation in 2019 and by in-kind support from our partners the Victorian Department of Transport. Our research partners also include the Australian Urban Observatory and The Australian Prevention Partnership Centre.

The aim of this project was to measure health benefits arising from walking or cycling (known collectively as active transport) based on estimates from strategic transport models and data provided by our Victorian Department of Transport partners. Active transport has been extensively studied and shown to be a significant factor for increasing levels of physical activity and health. However, quantification of health benefits from active transport is complex, which is why it is often missing from strategic transport planning, business cases and public dialogue, despite the general acceptance that more physical activity leads to long term health benefits.

Our partners and many others have specifically identified a gap in current approaches to measuring health benefits from active transport. Through this project we aimed to build capacity and understanding about the role of health and its measurement through the development of health impact assessment models and tools. A further aim of this project was to establish longer term relationships with policymakers and practitioners including the Victorian Department of Transport.



With a focus on research translation and capacity building, the project undertook several activities for measuring health benefits arising from active transport, including:

- a pre and post-project survey of our partners knowledge on Health Impact Assessments;
- a series of interviews of transport planners and practitioners to support the development of a case study application;
- a series of meetings between RMIT researchers and Victorian Department of Transport practitioners to learn about health impact assessment and transport modelling and for establishing a case study application;
- development of the *Transport Health Assessment Tool for Melbourne (THAT-Melbourne)*, a user friendly freely accessible web-based active transport health impact assessment tool housed within the [Australian Urban Observatory](http://aou.org.au) (aou.org.au);
- development of publications based on the model underlying *THAT-Melbourne* and on the process of research translation for the project.

The project has been successful in meeting its aims. The *THAT-Melbourne* tool has been disseminated to a wide audience through the [Australian Urban Observatory](http://aou.org.au) (aou.org.au) and has been accessed over 2,500 times at the time of writing. In 2021, the project also received a Victorian Planning Institute of Australia (PIA) Award for Planning Excellence in Planning Research.

In seeking to develop longer term relationships, the project has contributed to the development of a broader project that was awarded joint funding by the United Kingdom Research Institute and the National Health and Medical Research Council titled *Joining Impact models of transport to the Built Environment: JIBE*.

In *JIBE* we will continue to develop models of transport behaviour that link to health, expanding them to include environmental impacts such as air quality and road injuries. For *JIBE*, our Victorian Department of Transport partners will continue to work with us as part of our Project Advisory Group. Finally, the model underlying *THAT-Melbourne* is currently being expanded to include more chronic diseases to improve the accuracy of measuring health benefits that come from using transport modes that promote physical activity.



Introduction

Globally, systematic quantification of active transport related health and economic benefits is overlooked in strategic transport planning. This is due to both limitations in modelling and technical capabilities, and a lack of knowledge and capability amongst policymakers and practitioners. As new methods are developed there is a need to translate this knowledge back to those most likely to use it in their roles as policymakers and planners. Despite advancements in this area, there is a knowledge gap and limited consideration, of active transport health and economic benefits when planning and investing in city-wide infrastructure that supports active transport behaviour and up-take.

Why is this a problem?

Population growth and a continued reliance on motor vehicle transport are global issues causing a host of societal problems including congested roads, slow travel times, increased air quality issues, decreased work-life balance due to long commute times, and increased sedentary behaviour. Australian capital cities are typically car-oriented and only some areas are supported with safe cycling and pedestrian infrastructure. However, the active transport modes of walking, cycling and public transport use that remedy congested road networks struggle to keep pace with increases in population and levels of transport demand.

Active transport is viewed as a more sustainable and healthy option. Unlike cars, walking and cycling confer health benefits through physical activity and have no emissions that impact air quality. Public transport requires users to walk, making it a more sustainable and healthy transport option for mass transit. Furthermore, active transport and cities with supportive infrastructure align to global efforts for more sustainable, equitable and community focussed cities as envisaged in the United Nation's Sustainable Development Goals 3, 10, 11 and 13

(good health and wellbeing, reduced inequalities, sustainable cities and communities, and climate action) that many organizations are now seeking to follow. (1)

Evidence of the health impacts and economic benefits of sustainable development are required to incentivize the provision of safe infrastructure that is supportive of active transport and health. Yet, it is often missing or underestimated due to a lack of rigorous tools that are readily accessible for a broad audience of users. Consequently, the role of health is often omitted in the planning phases of major infrastructure processes.

Existing tools for measuring health impacts in the Northern European and American contexts include the Health Economic Assessment Tool (HEAT) and Integrated Transport and Health Impact Modelling Tool (ITHIM). Whilst, in Australia methods detailed in the Australian Transport Assessment and Planning (ATAP) guidelines have been used previously. (2) However, until now, there have been no readily available or accessible tools for calculating health impacts in the Australian context that use Australian health data, although models for this have been developed by researchers. (3)

This project sought to rectify this in a partnership setting with the Victorian Department of Transport (DOT). The interdisciplinary team of RMIT University researchers developed the tool, with scenario development and co-design from Department of Transport colleagues. The Transport Health Assessment Tool for Melbourne (THAT-Melbourne) sits within the Australian Urban Observatory (auo.org.au) and was disseminated to a broad audience of policymakers and practitioners since being launched in April 2021. Furthermore, the model underlying the tool and methods for evaluating health impacts have been further developed to assess health impacts from transport behaviours derived from DOT strategic models.

The remainder of this document provides details of the interdisciplinary research team from RMIT University and DOT and details the components of the project that were used to support development of the Transport Health Assessment Tool for Melbourne (THAT-Melbourne).



The Project Team

The project comprised of an interdisciplinary team of researchers and practitioners from RMIT University and the Victorian Department of Transport. Expertise for this project was in the areas of transport modelling, health impact assessment, geospatial science, planning, psychology, statistics, economics, and policy and knowledge translation.

RMIT University



Dr Lucy Gunn (RMIT Lead)
Senior Research Fellow



Dr Belen Zapata-Diomedí
Vice-Chancellor's
Research Fellow



Dr Alan Both
Research Fellow



**Associate Professor
Melanie Davern**
Director Australian
Urban Observatory



Dr Annette Kroen
Research Fellow



Dr Chris De Gruyter
Vice-Chancellor's
Research Fellow



Dr Geoff Browne
Research Fellow



Ms Mahsa Abdollahyar
Research Assistant

Victorian Department of Transport



Mr Hugh Batrouney
(DOT Lead) Assistant
Director, Transport
Analysis & Modelling



Mr Morteza Chalaki
Senior Economist
Transport Analysis
& Assessments



Ms Anh Nguyen
Transport Analysis
& Modelling



Mr Rick Williams
Manager, Demand
Forecasting

Our Funders and Research Partners

This research was funded by RMIT University's Enabling Capability Platform Opportunity Fund for Research Translation and Impact and the Victorian Department of Transport, who provided in-kind and modelling support. The project was further supported by the Australian Urban Observatory which houses THAT-Melbourne and The Australian Prevention Partnership Centre.



RMIT University is known for its applied, policy-relevant and collaborative approach of working with government and industry in undertaking innovative research. It has funded RMIT researchers from the Centre of Urban Research. RMIT University's Enabling Capability Platform connects researchers from across disciplines into eight primary research clusters to harness resources and expertise to the benefit of high-quality relevant research outcomes.

▼
More information can be found on the RMIT Enabling Capability website rmit.edu.au/research/our-research/enabling-capability-platforms



The Victorian Department of Transport plans and operates transport in Victoria, Australia and is responsible for the long-term planning and development of Victoria's transport network ensuring integration between differing modes including public transport, walking, cycling, and cars.

▼
More information can be found on the DOT website transport.vic.gov.au



The AUO is a digital platform that transforms complex urban data into easily understood liveability maps across Australia's 21 largest cities developed within the Centre of Urban Research at RMIT University. The Observatory provides observation, information, understanding and resources to support action including future investment, planning and policy to create equitable, healthy and liveable places.

▼
More information can be found on the Observatory website auo.org.au



The NHMRC funded Australian Prevention Partnership Centre (The Prevention Centre) is a national initiative aimed at preventing chronic and lifestyle related diseases through the co-production of prevention related research developed through a network of over 150 leading academics, policy makers and practitioners. The Prevention Centre takes a systems-based approach to tackling the chronic disease epidemic. Since its inception, it has funded RMIT researchers to develop liveability indicators and to conduct research on liveability, complex systems and the health impacts of creating healthy liveable cities.

▼
More information can be found on the Prevention Centre website preventioncentre.org.au

Project Overview

About the health impact assessment model

The foundation of the project rests upon the health impact assessment (HIA) model developed by RMIT researchers building on original work for physical activity interventions from the Assessing Cost Effectiveness in Prevention Project.^(3,4) The HIA model captures health impacts due to increases in physical activity associated with active transport for the Melbourne population. Health impacts include the reduction in new cases of diseases and deaths, and health benefits in the form of Health-Adjusted Life Years (HALYs)¹ and Life Years.² ⁽⁵⁾

Figure 1 Provides schematic representation of the modelling process for the HIA model that underlies the *Transport Health Assessment Tool for Melbourne (THAT-Melbourne)*

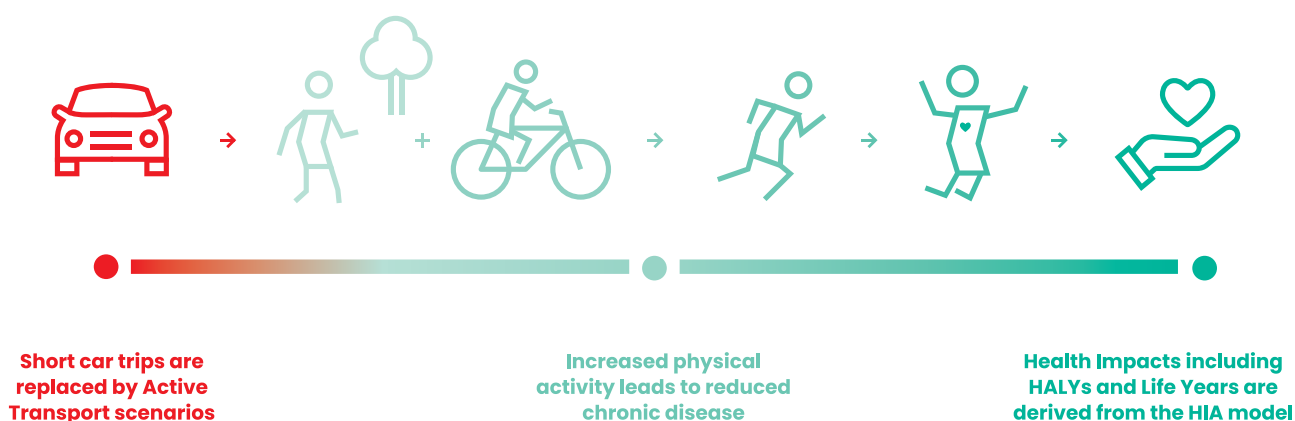


Figure 1: Overview of modelling framework

1. Health Adjusted Life Years (HALYs) are holistic measures of health that account for morbidity, mortality and quality of life.
2. Life Years are similar to HALYs however, they exclude the quality of life component.

Project Components

The project consisted of several concurrent components that supported the translation of the HIA model into an accessible tool. A summary of these components is detailed in the sections that follow and included:

- ▲ Workshops with the Victorian Department of Transport for case study development and application;
- ▲ Survey of policymakers and practitioners for scenario development;
- ▲ Development of the *Transport Health Assessment Tool for Melbourne (THAT-Melbourne)*;
- ▲ Online web-portal and User Interface development within the *Australian Urban Observatory*.

Workshops with Department of Transport

A series of five workshops were planned between RMIT and DOT colleagues. The first of these detailed the methods for the HIA model. This first workshop also surveyed workshop participants, to establish the extent of existing knowledge surrounding the measurement of health in transport modelling.

The second workshop was hosted by DOT and detailed information on their transport models. Detail focussed on the Victorian Integrated Transport Model (VITM) – a traditional four step model used for strategic transport planning purposes.

The remaining workshops were held as online meetings to accommodate working from home conditions resulting from the COVID19 pandemic. In these meetings a case study was developed and tested with the HIA model. The case study consisted of two scenarios: the first scenario explored the impact of more frequent train service and the second scenario added more frequent buses.

Methods were developed to connect the HIA model to the output from DOT's VITM model. This process built capacity in the RMIT research team.

A short report was provided to DOT detailing the methods in joining the two models. This report will act as support for the use of other bespoke transport models that more accurately capture the impacts of increased physical activity from active transport walking and cycling, such as Agent and Activity Based Models currently being developed RMIT researchers that are suitable for use with the HIA model.

07

Survey of policymakers and practitioners for scenario development

Throughout 2020, a series of interviews with 24 policy makers and practitioners working in the fields of active transport and transport planning were completed as part of this project for developing scenarios for *THAT-Melbourne*.

Interviews were conducted by Dr Lucy Gunn and PhD Student Afshin Jafari, whose PhD research focusses on cycling. The interviews were semi-structured with Afshin Jafari leading the questions on cycling, and Dr Lucy Gunn leading questions on walking and public transport. The aim of the interviews was to learn more about the gaps and priorities in city wide active transport infrastructure that could potentially support the development of models by

RMIT researchers. This had relevance for multiple projects including: The Australian Prevention Partnership Centre's funded project *Benchmarking, modelling and valuing the health liveable city*; and the National Health and Medical Council and United Kingdom Research Institute's funded project *JIBE: Joining Impact models of transport to the Built Environment*.

Only About Children

First

Transport Health Assessment Tool for Melbourne (*THAT-Melbourne*) and online web-portal

THAT-Melbourne is a freely accessible web-based simulation tool of transport and health housed within the [Australian Urban Observatory](#). It measures health impacts arising from physical activity when replacing short car trips with walking and cycling for transport. It uses the HIA model with a simple interface that allows users to select criteria for a variety of scenarios for walking, cycling or a combination of both. Each scenario can be tailored to accommodate different trip types, ages and sex of the population being modelled, as shown in Figure 2.

The tool aims to support knowledge translation, capacity building and infrastructure planning for active transport across transport, planning and teaching fields. For example, the health impacts calculated with the tool provides evidence for planning and teaching purposes and for advocating for infrastructure such as cycle lanes and footpaths that support healthy and active transport that can be used in reports and business cases.

An overview of the HIA model underlying *THAT-Melbourne* is presented in the next section.

Throughout 2021, RMIT researchers have presented *THAT-Melbourne* to a broad variety of audiences including policymakers, practitioners, and students both locally and globally, as detailed in the *Promotion and Dissemination* section.

Transport Health Assessment Tool

The Transport Health Assessment Tool for Melbourne (THAT-Melbourne) was developed as part of a larger research partnership project with the Victorian Department of Transport (DoT). The project was led by Dr Lucy Gunn, Dr Belen Zapata-Diomedí, Dr Alan Both, Dr Annette Kroen, Dr Chris De Gruyter and A/Prof Melanie Davern at RMIT University along with Hugh Batrouney, Dr Morteza Chalak, Anh Nguyen, and Rick Williams from the Victorian Department of Transport. The project was funded by an RMIT University Enabling Capability Platform Opportunity Fund on Research Translation and Impact and with in-kind support from the Victorian Department of Transport.

Scenario: replacing car trips under 1km with walking for all trip purposes

Trips to replace:
☒ All ?
☐ Commuting ?

Replace car trips with:
 Walking
☒ 0 – 1km with walking
☐ 0 – 2km with walking

Sex
☒ All ☐ Female ☐ Male

Age Group²
 All age groups

Description Incidence Mortality Health Download

This scenario shows the results of replacing car trips under 1km for leisure, shopping, work, education or other purposes with walking trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 26.7% to 29.8% and from 62.9% to 59.8% for car trips taken as either a driver or passenger.

Increases in walking translate into a shift from 53.5% to 54.9% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.

Trip Mode	Base case (%)	Scenario (%)
Other	0.7%	0.7%
Driving	62.9%	59.8%
Public transport	8.4%	8.4%
Cycling	1.3%	1.3%
Walking	26.7%	29.8%

Figure 1: Distribution of base case and scenario trips.

Figure 2: Screenshot showing web access to the Transport Health Assessment Tool for Melbourne

THAT-Melbourne Model

The HIA model underlying *THAT-Melbourne* is a hybrid consisting of three sections:

01



Scenarios for replacing short car trips by walking, cycling or a combination of both walking and cycling modifiable by trip type, sex and age;

02



A measure of the proportional change in chronic disease risk due to an increase in physical activity from active transport; and

03



A proportional multi-state life table (PMSLT) that captures health impacts due to reduction in chronic diseases due to increased physical activity.

The scenarios and proportional change in chronic disease risk are a micro simulation at the individual level and the PMSLT calculations are a macro simulation aggregated by age and sex groups.

Figure 3 provides detail on the analytical framework for *THAT-Melbourne*. For interested readers a more detailed explanation can be found in the [Technical Report \(6\)](#) on the [Australian Urban Observatory](#) website.

Figure 3: Analytical framework for the Transport Health Assessment Tool for Melbourne

THAT-Melbourne has a series of scenarios defined for the replacement of short car trips with walking, cycling or a combination of both over short distances of 1 km, 2 km, 5 km, or up to 10 km with downloadable reports for these scenarios available on [FigShare](#). (7)

The scenarios can be adjusted using the following criteria:

- All trip types (includes leisure, shopping, work, education or trips for other purposes);
- Commuting trips (includes trips for work and educational purposes only);
- Sex (male, female, or both);
- Age in years (16-19, 20-29, 40-64, 65 plus, or all age groups).

The health impacts contained within the tool relate to seven chronic diseases that are impacted by physical activity which include uterine cancer, lung cancer, stroke, ischemic heart disease, diabetes type 2, colon cancer and breast cancer.

The health impacts presented by the tool include changes in *incidence* as the rate of new cases of these diseases, *mortality* as the rate of deaths from these diseases, and *Health Adjusted Life Years* (HALYs) which combines gains in both length and quality of life. The tool provides graphs, metrics, and explanations for changes in mortality, morbidity, and HALYs that can be used for reporting and advocacy purposes, example outputs from the [Australian Urban Observatory](#) and *THAT-Melbourne* are shown in Figures 4, 5, & 6.

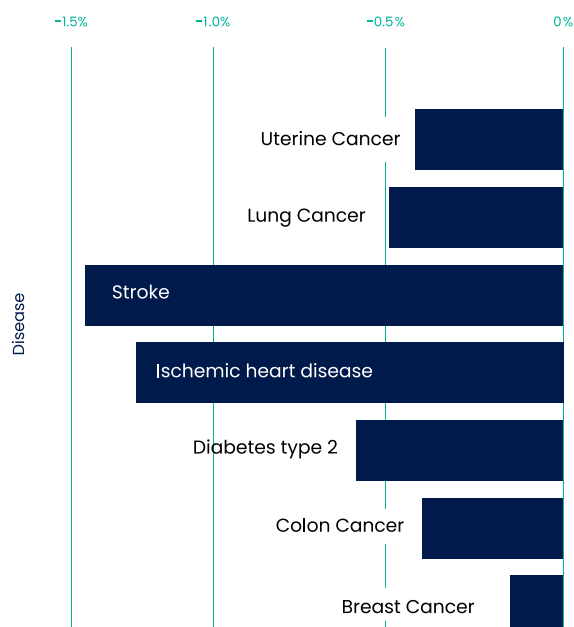


Figure 4: Graph showing percentage change in incidence by disease

HALYS

The model estimates a total of **75,740** HALYs for the scenario population.

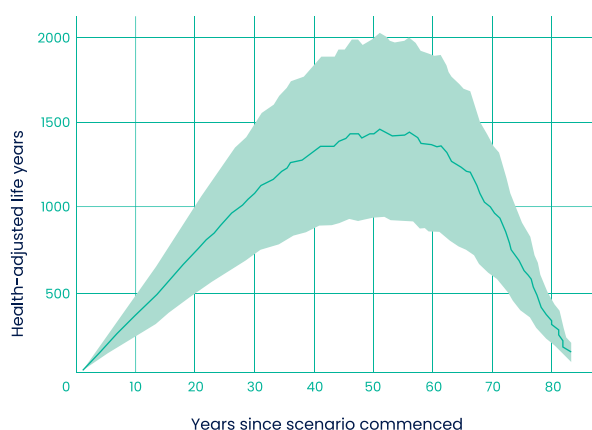


Figure 5: Graph showing Health-Adjusted Life Years (HALYs)

Percentage change in mortality by disease

In this scenario, Figure 4 shows that mortality is reduced by:

0.20%	for colon cancer
0.49%	for diabetes type 2
0.98%	for ischemic heart disease
1.05%	for stroke
0.37%	for lung cancer
0.26%	for uterine cancer
0.04%	for breast cancer

This translates into the following total number of prevented deaths aggregated across the years as shown in Figure 5.

46	cases of colon cancer
22	cases of diabetes type 2
5,355	cases of ischemic heart disease
2,430	cases of stroke
511	cases of lung cancer
21	cases of uterine cancer
14	cases of breast cancer

Figure 6: Estimates of health impacts in terms of mortality

THAT-Melbourne Launch

THAT-Melbourne was launched in April 2021 through a webinar to an audience predominantly consisting of government and industry policymakers and practitioners with over one hundred people in attendance.

The webinar was chaired by Dr Lucy Gunn and opened by current Enabling Capability Platform Director Professor Lauren Rickards representing the RMIT Capability Platform and RMIT funder who officially launched the tool as part of the session. This was followed by a lively presentation from Dr Patrick Harris, Senior Research Fellow at University of New South Wales, Sydney on *The Importance of measuring health to advance policies and dialogue*.

Presentations were given by Dr Morteza Chalak, Senior Economist, Transport Analysis and Assessments of the Victorian Department of Transport, and by the modelling leads Drs Belen Zapata-Diomedes and Alan Both who also launched the tool and presented a technical overview of the modelling and summary of how to use THAT-Melbourne.

The Director of the Australian Urban Observatory, Associate Professor Melanie Davern introduced the Observatory web-portal housing THAT-Melbourne and presented an overview of the website capabilities and liveability indicator data covering Australia's 21 largest cities.

Since the tool's launch, it has become one of the Australian Urban Observatory's most frequently visited pages and has been accessed over 2,500 times at the time of writing.

09

Planning Institute of Australia Award

Knowledge translation and capacity building were primary aims of this project. *THAT-Melbourne* allows practitioners to provide additional evidence for advocating for active transport infrastructure that if implemented will have a lasting impact on how we plan and build cities to be more healthy, sustainable and liveable. In 2021, RMIT researchers nominated *THAT-Melbourne* for a Planning Institute of Australia (PIA) Award for Excellence in Planning Research because of the tool's potential influence on planning. The PIA Awards recognize excellence in planning across a range of disciplines

and sectors seeking out the best of planning state-wide and nationally.

The RMIT research team won the 2021 Award for Planning Excellence in Victoria for Planning Research announced at the Annual PIA Christmas Party and Award Ceremony in December 2021, with team members from the Healthy, Liveable Cities Lab also receiving a Commendation for their liveability research. *THAT-Melbourne* will now go through to the PIA National Awards.

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Promotion and dissemination

Since the tool's launch, the RMIT research team has undertaken several presentations on Health Impact Assessments as part of the RMIT short course on *Integrating Health and Planning* and to RMIT Europe as part of the European Institute of Innovation and Technology (EIT) and RMIT Europe Urban Mobility series on health and the environment and as part of the Massive Online Open Course on liveability and mobility. Lead modeller, Dr Belen Zapata-Diomedes has presented on *THAT-Melbourne* four times, including at the 2021 State of Australian Cities conference in Melbourne, where she was awarded a prize for best early career researcher presentation under the City Health and Liveability theme.

In further promoting the tool, RMIT researchers wrote an article titled *Helping planners understand health benefits through the Transport Health Assessment Tool for Melbourne (THAT-Melbourne)*³, published in the PIA industry journal, Planning News

which has a targeted audience of those from the planning field.

As part of the project, two academic publications are currently being drafted. The first focuses on the technical modelling side, whilst the second is providing an overview and analysis of the knowledge and the research translation process drawing on different information captured from the surveys of policymakers and practitioners undertaken throughout the project. It is anticipated that both publications will be completed and ready for dissemination in 2022.

3. Gunn, L, Davern, M, Zapata-Diomedes, B, Both, A, Kroen, A, De Gruyter, C (2021). Helping planners understand health benefits through the Transport Health Assessment Tool for Melbourne (THAT-Melbourne), Planning News, Vol. 47, No. 5, June 2021

What's Next?

The role of health in our society could not be clearer as demonstrated by the challenges faced by many throughout the current COVID19 pandemic. Despite this, the complexity of measuring health accurately using robust context specific tools remains difficult. *THAT-Melbourne* fills this gap and provides a tool for measuring health benefits from physical activity accurately for use by policymakers, practitioners and general users for a Melbourne population. However, the models underlying these tools are complex and their development takes time. The current version of the model provides health impacts across a lifetime for one generation of people. This model is yet to account for dynamic population growth from which we could expect even greater projected population health impacts. Hence, this project is foundational with future work seeking to include new additions to improve the accuracy of the health impact assessment model. Furthermore, it is possible to extend the modelling work to other Australian or international cities contingent on health and transport data availability and partnership opportunities.

The current model focuses on seven chronic diseases related to physical activity but there are many more health impacts that relate to physical activity and mental health that could be included. Furthermore, walking and cycling are not without injury risk due to on-road collisions. The removal of motor vehicles undertaking short trips has additional benefits to other road users by not adding to congestion or the environment due to reductions in vehicle related green-house gases and pollutants. Both of these impacts could be included in future HIA modelling to expand the current version of *THAT-Melbourne*.

In summary, estimating health impacts accurately requires transport models in addition to HIA models and two RMIT University projects and their research collaborators, including our DOT partners, are seeking to address these remaining gaps and modelling complexities. We describe these projects and their progress briefly in the sections that follow.



Agent and Activity Based Models

Agent and Activity Based Models are currently being developed as part of the NHMRC funded Australian Prevention Partnership Centre's project *Benchmarking, monitoring and valuing the healthy, liveable city* being undertaken by RMIT researchers. These models simulate travel behaviour based on the characteristics of real people and their travel itineraries as derived from Department of Transport (DOT) data from the Victorian Integrated Survey of Travel and Activity (VISTA). These models accurately show how travel behaviour differs between transport scenarios. Specifically, these models, for example, are able to show the difference in walking distances and physical activity due to major infrastructure changes. Measuring incremental changes to physical activity is a key input for the HIA model developed here, hence these Agent and Activity Based Models have the potential to be combined with the HIA model. Currently, the RMIT research team is developing methods to join these models together to measure health impacts more accurately from hypothetical or planned changes to infrastructure.

The JIBE Project

Our partnership with Department of Transport (DOT) enabled RMIT researchers to apply and secure over \$800,000 in NHMRC and UKRI grant funding for a joint project measuring transport demand and health impacts concurrently within one modelling framework. The *Joining Impact models of transport to the Built Environment: JIBE project* is drawing on more detailed methods that accommodate road injuries, air pollution and additional exposure pathways for chronic diseases to more accurately measure both transport behaviour and subsequent health impacts.

This project is being jointly led by researchers from RMIT University and from The University of Cambridge and involves several other leading Universities including Monash University, The University of Melbourne, The University of Queensland, London School of Hygiene and Tropical Medicine, Imperial College London, and The University of Leicester.

This project has two advisory groups, one in the UK and the other here in Melbourne that includes members from transport and planning related fields helping to expand our research translation reach discussed further in the next section.

Capacity Building and Research Translation

A final aspect of this project has been the emphasis on building capacity and on research translation. Health is an important aspect of any society. Yet, as discussed throughout this report, measuring health impacts is challenging to do in practice due to the complexity of linking behaviour to health outcomes. Furthermore, there are limited tools in existence that make it easy for policy makers, practitioners, industry and the community to access for evidence linking transport to health impacts. People wanting to measure health impacts accurately may also lack the knowledge or expertise to do this confidently. Hence, building skills within our research team for measuring health impacts and building capacity by teaching others about our methods and results has been a large component of this project.

This was a partnership research project and as researchers, we have learnt much from our Department of Transport (DOT) partners and webinar participants about modelling transport demand and behaviours. These are key inputs for measuring changes in behaviour and physical activity that are influenced by infrastructure changes, which can be measured through case studies or scenarios.

Building capacity between researchers and both government and industry-based colleagues through partnership projects like this encourages routine uptake and use of tools for measuring health impacts and also informs our future research. This is the process of research translation. Research translation is not one-way but is bi-directional between researchers and partners. It also involves a variety of activities that seek to improve and influence knowledge and use of research in practice. This project is a great example of policy-focused research co-production with government partners that aligns closely with the RMIT research strategy of working in partnership with industry to solve real world problems.

Since the tool has been launched, the team has received inter-state enquiries regarding development of similar tools and research translation continues. The success of this project is demonstrated through the collaborative and productive research partnership established between research and policy partners with the influence of the tool on prioritising and planning active transport infrastructure necessary for a healthy, sustainable and liveable city.

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